

NUMBER SYSTEM – APTITUDE LEARNING NOTES

Complete concepts • formulas • shortcuts • solved examples • practice

1. Number System Basics

Natural numbers (N): 1, 2, 3, ... (some conventions include 0). **Whole numbers:** 0, 1, 2, ... **Integers (Z):** ..., -2, -1, 0, 1, 2, ... **Rational:** p/q , $q \neq 0$. **Irrational:** cannot be written as p/q , e.g. $\sqrt{2}$, π . **Real:** rational + irrational.

Positive/negative: Positive numbers are >0 ; negative numbers are <0 . **Prime:** integer greater than 1 with exactly two positive divisors. **Composite:** integer greater than 1 with more than two divisors. 1 is neither prime nor composite.

Even: divisible by 2. **Odd:** not divisible by 2. Consecutive integers differ by 1; consecutive even/odd integers differ by 2.

2. Place Value, Face Value & Expanded Form

For 5,274: face value of 7 = 7; place value of 7 = 70. Expanded form = $5 \times 1000 + 2 \times 100 + 7 \times 10 + 4$.

Number of digits: For positive integer n , digits = $\text{floor}(\log_{10} n) + 1$. For $10^{(k-1)} \leq n < 10^k$, n has k digits.

3. Divisibility Rules

Divisor	Rule
2	Last digit is 0,2,4,6,8.
3	Sum of digits is divisible by 3.
4	Last 2 digits are divisible by 4.
5	Last digit is 0 or 5.
6	Divisible by both 2 and 3.
7	Double last digit and subtract from remaining number; repeat.
8	Last 3 digits are divisible by 8.
9	Sum of digits is divisible by 9.
10	Last digit is 0.
11	Difference between alternating digit sums is 0 or a multiple of 11.
12	Divisible by both 3 and 4.
15	Divisible by both 3 and 5.
16	Last 4 digits are divisible by 16.
18	Divisible by both 2 and 9.
20	Last 2 digits divisible by 20.
25	Last 2 digits: 00,25,50,75.
100	Last 2 digits are 00.
125	Last 3 digits divisible by 125.

4. Factors, Multiples, Prime Numbers

Factor: a number that divides another exactly. **Multiple:** result of multiplying a number by an integer. Every positive integer has a finite number of factors but infinitely many positive multiples.

Prime factorization: $360 = 2^3 \times 3^2 \times 5$. **Fundamental theorem:** every integer greater than 1 has a unique prime factorization apart from order.

Number of factors: If $n = p_1^a \times p_2^b \times \dots \times p_k^c$, then $d(n) = (a+1)(b+1)\dots(c+1)$. Example: $360 = 2^3 \times 3^2 \times 5^1 \rightarrow d=4 \times 3 \times 2=24$.

Sum of factors: $\sigma(n) = [(p_1^{a+1}-1)/(p_1-1)] \times \dots$. Example: for $12=2^2 \times 3$, $\sigma(12)=(1+2+4)(1+3)=28$.

Perfect square: every prime exponent is even. **Perfect cube:** every prime exponent is a multiple of 3.

5. HCF / GCD and LCM

HCF (GCD): greatest common divisor. **LCM:** least common multiple. For two positive integers a, b : $\text{HCF} \times \text{LCM} = a \times b$.

Prime method: HCF uses minimum exponents; LCM uses maximum exponents. Example: $72=2^3 \times 3^2$ and $120=2^3 \times 3 \times 5 \rightarrow \text{HCF}=2^3 \times 3=24$, $\text{LCM}=2^3 \times 3^2 \times 5=360$.

Euclidean algorithm: $\text{gcd}(a,b)=\text{gcd}(b, a \bmod b)$. Example: $\text{gcd}(252,105)=\text{gcd}(105,42)=\text{gcd}(42,21)=21$.

Useful: If numbers leave the same remainder r when divided by d , then d divides each pairwise difference. Greatest such divisor is gcd of the differences.

6. Remainders & Modular Arithmetic

If $a = bq + r$, then $0 \leq r < b$. In modular notation, $a \equiv r \pmod{b}$.

Rules: $(a+b) \bmod m = [(a \bmod m) + (b \bmod m)] \bmod m$; similarly for subtraction and multiplication. For powers, reduce the base first.

Example: $7^2 \bmod 5$. Since $7 \equiv 2 \pmod{5}$, calculate 2^2 . Because $2 \equiv 1 \pmod{5}$, $2^2 \equiv (2 \bmod 5)^2 \equiv 1$.

Negative remainder: adjust to the standard non-negative remainder. $-3 \bmod 5 = 2$.

7. Unit Digit / Last Digit

For powers, look for a repeating cycle in the unit digit. Typical cycles: $2 \rightarrow 2, 4, 8, 6$ (length 4); $3 \rightarrow 3, 9, 7, 1$; $7 \rightarrow 7, 9, 3, 1$; $8 \rightarrow 8, 4, 2, 6$; $4 \rightarrow 4, 6$; $9 \rightarrow 9, 1$; $5 \rightarrow 5$; $6 \rightarrow 6$.

Method: If cycle length is k , divide exponent by k . Use exponent remainder; if remainder is 0, use the k th cycle value.

Example: unit digit of 3^{47} : $47 \bmod 4 = 3 \rightarrow$ third value in $3, 9, 7, 1 \rightarrow 7$.

8. Last Two / Three Digits

To find the last 2 digits, work modulo 100; last 3 digits, modulo 1000. Reduce after each multiplication to keep numbers small.

Example: last two digits of $7^4 = 2401 \rightarrow 01$. For larger powers, repeatedly square modulo 100.

9. Fractions & Decimals

Terminating decimal: In lowest terms p/q , decimal terminates iff q has no prime factors other than 2 and 5. Example: $7/40$ terminates because $40=2^3 \times 5$.

Recurring decimal: $1/3=0.333\dots$; $2/11=0.1818\dots$. For a fraction p/q with q coprime to 10, the repeating length is related to the smallest k for which $10^k \equiv 1 \pmod{q}$.

Fraction comparison: cross-multiply positive fractions: a/b vs $c/d \rightarrow$ compare ad and bc .

10. Ratio, Proportion & Number Relations

$a:b = c:d$ means $ad=bc$. If $a:b=m:n$, let $a=mk$ and $b=nk$. This substitution is extremely useful in aptitude problems.

Mean proportional: if $a:x=x:b$, then $x=\sqrt{(ab)}$. **Third proportional:** $a:b=b:x \rightarrow x=b^2/a$.

11. Successive Numbers & Averages

Average = sum / number of items. Sum of first n natural numbers = $n(n+1)/2$. Sum of first n odd numbers = n^2 . Sum of first n even numbers = $n(n+1)$.

Sum of squares = $n(n+1)(2n+1)/6$. Sum of cubes = $[n(n+1)/2]^2$.

Consecutive integers: $n, n+1, \dots, n+k-1$. Their average is the middle term when the count is odd; otherwise the average is the mean of the two middle terms.

12. Algebraic Number Identities

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\text{If } a+b+c=0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

13. Divisibility & Digit-Sum Tricks

For 9 and 3, digit sum is the fastest test. For 11, alternate-sum test is useful. For 6, check 2 and 3. For 12, check 3 and 4. For 15, check 3 and 5.

Digital root: repeatedly add digits until one digit remains. For positive n , digital root is $n \bmod 9$, with remainder 0 represented as 9. Example: $98765 \rightarrow 9+8+7+6+5=35 \rightarrow 8$.

14. Factorials & Trailing Zeros

$n! = 1 \times 2 \times \dots \times n$. Trailing zeros come from factors of $10 = 2 \times 5$. Since factorials contain more 2s than 5s, count the number of 5s.

Zeros in $n!$: $\text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$. Example: $100! \rightarrow 20+4=24$ zeros.

Highest power of prime p dividing $n!$: $\text{floor}(n/p) + \text{floor}(n/p^2) + \text{floor}(n/p^3) + \dots$.

15. Highest Power / Number of Factors in a Product

To find the highest power of p dividing n , first prime-factorize n . For a product, add the exponents of p from every factor. For $n!$, use Legendre's formula above.

Example: exponent of 2 in $10! = 5+2+1=8$.

16. Perfect Squares & Cubes

A perfect square cannot have an odd exponent in prime factorization. A perfect square's unit digit can only be 0,1,4,5,6,9. It can never end in 2,3,7,8.

A perfect cube can end in any digit. To make a number a perfect square, multiply by the missing prime factors needed to make every exponent even. For a cube, make every exponent a multiple of 3.

17. Counting Numbers with Conditions

Multiples of k from 1 to n : $\text{floor}(n/k)$. **Multiples of k from a to b :**
 $\text{floor}(b/k) - \text{floor}((a-1)/k)$.

Multiples of at least one of a, b : $\text{floor}(n/a) + \text{floor}(n/b) - \text{floor}(n/\text{lcm}(a,b))$. This is inclusion-exclusion.

Numbers divisible by neither a nor b : total – divisible by a or b .

18. Base Systems

In base b , digits range from 0 to $b-1$. Decimal value of $(d_{n-1} \dots d_1 d_0)_b$ is $\sum d_i b^i$. Binary uses base 2; octal base 8; hexadecimal base 16.

Binary $\rightarrow$ **decimal**: $101101_2 = 32+8+4+1 = 45$. **Decimal** $\rightarrow$ **binary**: repeatedly divide by 2 and read remainders upward.

Shortcut: 1 hex digit = 4 binary bits; 1 octal digit = 3 binary bits.

19. Common Aptitude Patterns

Pattern	Approach
Remainder pattern	Use modulo arithmetic; reduce bases and exponents using cycles.
Unit digit	Find cycle length of the last digit.
HCF/LCM	Prime factorize or use Euclidean algorithm.
Same remainder	Take HCF of pairwise differences.
Largest number leaving same remainder	Usually HCF of differences.
Smallest number divisible by several numbers	LCM
Smallest number to add/subtract for divisibility	Compute the current remainder.
Trailing zeros	Count factors of 5 in factorial/product.
Number of factors	Multiply (exponent+1) for each prime.
Perfect square/cube	Check prime exponents.
Digit divisibility	Use 3, 9, 11 and last-digit rules.
Counting multiples	Use floor division and inclusion-exclusion.

20. Solved Examples

Q1. Find HCF and LCM of 84 and 126.

Solution: $84=2^2 \times 3 \times 7$; $126=2 \times 3^2 \times 7 \rightarrow \text{HCF}=2 \times 3 \times 7=42$; $\text{LCM}=2^2 \times 3^2 \times 7=252$.

Q2. How many factors does 720 have?

Solution: $720=2^4 \times 3^2 \times 5 \rightarrow (4+1)(2+1)(1+1)=30$.

Q3. Find the remainder when 2^{100} is divided by 7.

Solution: $2^3=8 \equiv 1 \pmod{7}$. $100 \bmod 3=1 \rightarrow$ remainder 2.

Q4. Unit digit of 7^{123} .

Solution: Cycle 7,9,3,1. $123 \bmod 4=3 \rightarrow 3$.

Q5. Trailing zeros in $125!$.

Solution: $\text{floor}(125/5)+\text{floor}(125/25)+\text{floor}(125/125)=25+5+1=31$.

Q6. Greatest number that divides 43, 91 and 139 leaving the same remainder.

Solution: Take differences: $91-43=48$; $139-91=48 \rightarrow \text{gcd}=48$.

Q7. Smallest number divisible by 12, 15 and 18.

Solution: $12=2^2 \times 3$; $15=3 \times 5$; $18=2 \times 3^2 \rightarrow \text{LCM}=2^2 \times 3^2 \times 5=180$.

Q8. Is 123456 divisible by 3 and 9?

Solution: Digit sum=21. Divisible by 3, but not by 9.

Q9. Number of integers from 1 to 500 divisible by 6 but not 10.

Solution: Multiples of 6=83. Multiples of $\text{lcm}(6,10)=30$: 16. Answer=83-16=67.

Q10. Make 180 a perfect square using the smallest multiplier.

Solution: $180 = 2^2 \times 3^2 \times 5$. Only 5 has odd exponent $\rightarrow$ multiply by 5; $900 = 30^2$.

21. Practice Questions

1. Find the HCF of 96 and 144.
2. Find the LCM of 24, 36 and 54.
3. How many factors does 840 have?
4. Find the sum of all positive divisors of 72.
5. Find the unit digit of 9^{2026} .
6. Find the remainder when 5^{73} is divided by 6.
7. How many trailing zeros are in $200!$?
8. Find the greatest number that divides 65, 117 and 169 leaving the same remainder.
9. How many numbers from 1 to 1000 are divisible by 7?
10. How many numbers from 1 to 500 are divisible by 4 or 6?
11. Find the smallest number that must be added to 9876 to make it divisible by 11.
12. Check whether 123456789 is divisible by 9.
13. Find the least number by which 540 must be multiplied to become a perfect square.
14. Find the least number by which 675 must be multiplied to become a perfect cube.
15. Convert 1011011₂ to decimal.
16. Find the last two digits of 3^{20} .
17. Find the average of the first 50 natural numbers.
18. If two numbers have HCF 12 and LCM 420 and one number is 60, find the other.
19. Find the number of integers from 1 to 1000 divisible by neither 3 nor 5.
20. Find the highest power of 2 dividing $100!$.

22. Answer Key

- 1) 48 • 2) 216 • 3) 32 • 4) 195 • 5) 1 • 6) 5 • 7) 49 • 8) 52 • 9) 142 • 10) 167 • 11) 3 • 12) Yes • 13) 15 • 14) 25 • 15) 91 • 16) 01 • 17) 25.5 • 18) 84 • 19) 533 • 20) 97

23. Quick Revision Sheet

- HCF = product of common primes with minimum exponents.
- LCM = product of all primes with maximum exponents.
- $\text{HCF} \times \text{LCM} = \text{product of two numbers}$.
- Number of divisors of $p^a \times q^b \times \dots = (a+1)(b+1)\dots$
- Sum of first n numbers $= \frac{n(n+1)}{2}$.
- Sum of first n odd numbers $= n^2$.
- Sum of first n even numbers $= n(n+1)$.
- Trailing zeros in $n! = \text{floor}(\frac{n}{5}) + \text{floor}(\frac{n}{25}) + \dots$

- Multiples of $k \leq n = \text{floor}(n/k)$.
- Same remainder $\rightarrow$ use differences and their HCF.
- Last digit $\rightarrow$ cyclic pattern.
- Last 2 digits $\rightarrow \text{mod } 100$; last 3 $\rightarrow \text{mod } 1000$.
- Divisible by 3/9 $\rightarrow$ digit sum.
- Divisible by 11 $\rightarrow$ alternating digit-sum difference.
- Terminating $p/q \rightarrow$ reduced denominator contains only 2s and 5s.

Study strategy: Learn divisibility $\rightarrow$ prime factorization $\rightarrow$ HCF/LCM $\rightarrow$ remainders $\rightarrow$ unit digits $\rightarrow$ factorials/trailing zeros $\rightarrow$ counting multiples $\rightarrow$ mixed practice. For placement aptitude, practice timed sets after mastering each concept.